Assignment 1

Classical Machine Learning Methods

Cairo University

Faculty of Engineering – EECE Department

Course: Neural Networks (ELC4028)

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Derivation of the Perceptron Learning Rule

Consider a perceptron with input vector $\mathbf{x}$, weight vector $\mathbf{w}$, and bias b. The perceptron computes

$$a = \mathbf{w}^T\mathbf{x} + b, \tag{1}$$

and the predicted output is

$$\hat{y} = \text{sign}(a), \tag{2}$$

where the target label is $y \in \{-1, +1\}$.

Classification condition

A sample is correctly classified if

$$y(\mathbf{w}^T\mathbf{x} + b) > 0, \tag{3}$$

and misclassified if

$$y(\mathbf{w}^T\mathbf{x} + b) \leq 0. \tag{4}$$

Objective of learning

When a sample is misclassified, the goal is to update the parameters so that the quantity

$$y(\mathbf{w}^T\mathbf{x} + b)$$

increases, which moves the prediction toward the correct class.

Derivation of the update rule

To correct a misclassification, we need to increase the value of $y(\mathbf{w}^T\mathbf{x} + b)$.

A simple way to achieve this is to adjust the weights in the direction of the input vector. Since the direction depends on the label y, we choose the update to be proportional to $y\mathbf{x}$.

Thus, we propose the update:

$$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta y\mathbf{x}, \tag{5}$$

$$b_{new} = b_{old} + \eta y, \tag{6}$$

where $\eta > 0$ is the learning rate.

Proof that the update improves classification

Substitute the updated values:

$$\mathbf{w}_{new} = \mathbf{w} + \eta y\mathbf{x}, \quad b_{new} = b + \eta y.$$

Compute the new activation:

$$a_{new} = \mathbf{w}_{new}^T\mathbf{x} + b_{new}.$$

Substitute:

$$a_{new} = (\mathbf{w} + \eta y\mathbf{x})^T\mathbf{x} + (b + \eta y).$$

Expand:

$$a_{new} = (\mathbf{w}^T\mathbf{x} + b) + \eta y(\mathbf{x}^T\mathbf{x} + 1). \tag{7}$$

Multiply by y:

$$ya_{new} = y(\mathbf{w}^T\mathbf{x} + b) + \eta(\mathbf{x}^T\mathbf{x} + 1). \tag{8}$$

Since $\eta > 0$ and $\mathbf{x}^T\mathbf{x} \geq 0$, we have

$$\eta(\mathbf{x}^T\mathbf{x} + 1) > 0.$$

Therefore,

$$y(\mathbf{w}_{new}^T\mathbf{x} + b_{new}) > y(\mathbf{w}^T\mathbf{x} + b), \tag{9}$$

which shows that the update increases the value and moves the prediction toward correct classification.

Final learning rule

If

$$y(\mathbf{w}^T\mathbf{x} + b) \leq 0, \tag{10}$$

then update using

$$\mathbf{w} \leftarrow \mathbf{w} + \eta y\mathbf{x}, \quad b \leftarrow b + \eta y. \tag{11}$$

Otherwise, no update is performed.

Conclusion

The perceptron learning rule is an error-correction rule. It updates the parameters only when a sample is misclassified, and the update direction ensures movement toward correct classification.